

Non-Parametric Statistical Techniques

STA2020

Lecture 1: Introduction

Learning Outcomes

- Understand nonparametric techniques
- Distinguish parametric vs nonparametric
- Classify data types
- Rank data (including ties)

Parametric Techniques

- Assume known underlying distribution (usually normal)
- Use CLT for large samples
- Assumptions -> sampling distribution of test statistics were derived -> inferences about unknown pop parameters
- Focus on parameters
- Typically require quantitative data

Examples

- Estimating population mean μ when population SD σ is unknown -> use sample mean $\bar{x}$ and sample SD s -> t-distribution for inference
- Testing $H_0 : \mu = \mu_0$ -> use t-test with test statistic $t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$
- both rely on assumptions of normality!

Non-Parametric Techniques

- Make weak assumptions about underlying distributions (e.g. symmetry)
- Do not depend on parameters specific to a distribution.
- Use ranks, signs, frequencies
- Require independence and random sampling
- Most do not depend on the actual numerical values -> ranks!

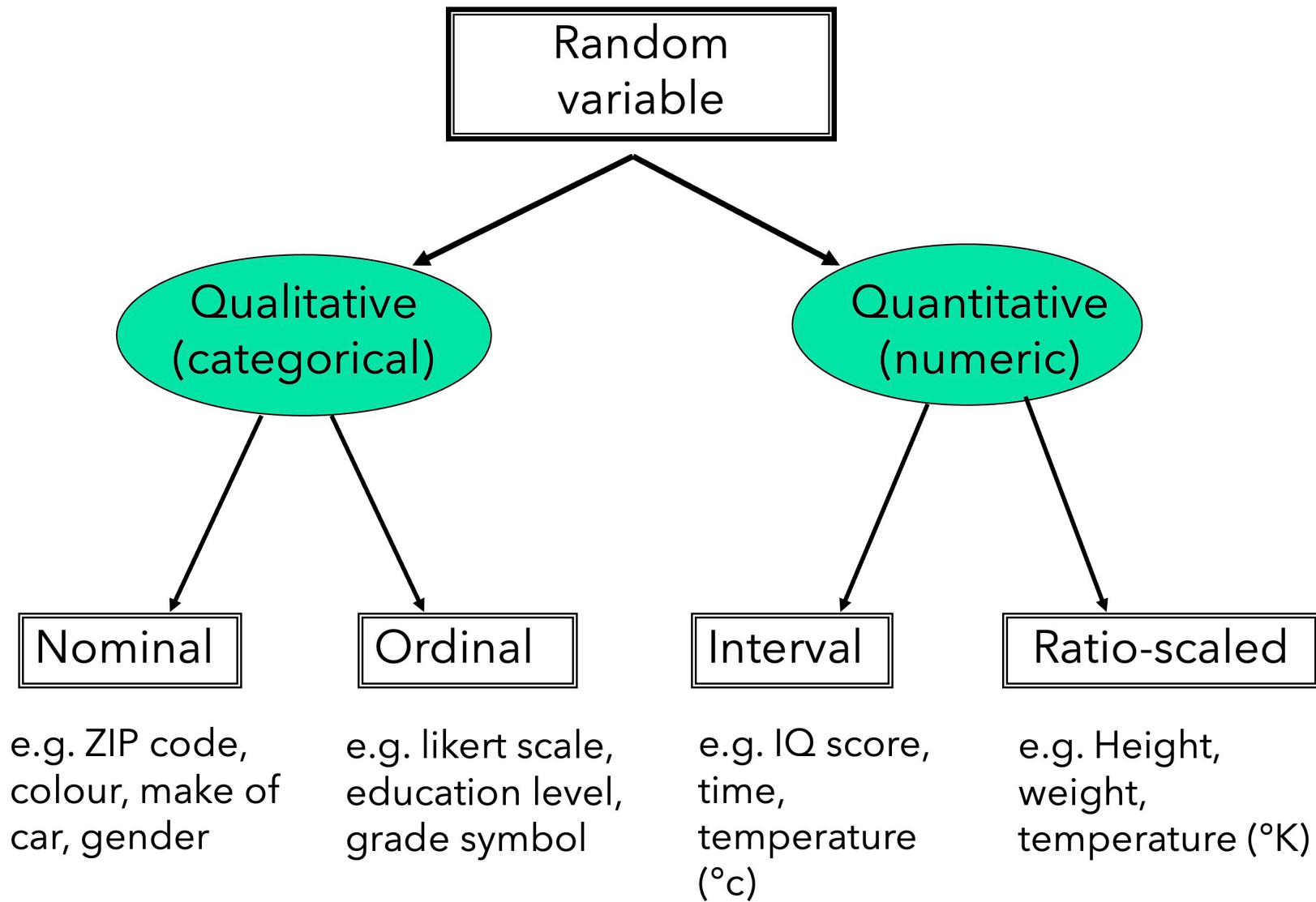
When to Use Nonparametric Tests

- Non-normal (or distribution is uncertain) quantitative data
- Small samples
- Qualitative data: ordinal or nominal data

Important note

- Nonparametric techniques are not restricted to non-normal data.
- However, in such cases, parametric techniques are preferred
- More power since we know the appropriate model
- Power? The probability of correctly rejecting a false null hypothesis (i.e. avoiding a Type II error).
- NP tests are always valid, sometimes not optimal in power.

Data Types



Data types

- Nominal: categories without order (e.g. blood type)
- Ordinal: categories with order (e.g. Likert scale)
 - can be ranked with respect to each other
 - but differences between categories are not necessarily meaningful
- Interval: numerical values with meaningful differences but no true zero (e.g. temperature in Celsius)
 - ratios between values are not meaningful (e.g. 20°C is not twice as hot as 10°C)
- Ratio: numerical values with meaningful differences and a true zero (e.g. weight)
 - ratios between values are meaningful (e.g. 20 kg is twice as heavy as 10 kg)

Interval vs Ratio

- Is clock time an interval or ratio variable?
- Time has a natural order
- Differences between points in time are equal, so difference between 12pm and 1pm is the same as 4pm and 5pm (i.e. one hour).
- BUT time does not have a meaningful zero, there is no zero time point (cyclical)
- So can't say 2pm is twice as "old of a time" than 1pm – NOT MEANINGFUL!
- Ratio is not meaningful!

Interval vs Ratio

- What about duration or length of time, e.g. seconds?
- There is a natural order, so 1 second is smaller than 2 seconds
- Equal distances between points, so the difference between 10 and 15 seconds is the same as between 5 and 10 seconds.
- AND it has a true zero, 0 seconds is meaningful and thus, so are ratios (10 seconds is twice as long as 5 seconds)
- Go through the same reasoning for temperature in Celsius vs in Kelvin

Examples

1. The number of students in a statistics class
2. The make of car driven by each of a sample of executives
3. The rating (Extremely poor[1], Very Poor[2], Poor[3], Unsure[4], Good[5], Very Good[6], Excellent[7]) reported for a particular television program by each of a sample of viewers
4. The weekly closing price of gold throughout the year
5. The month of highest sales for each firm in a sample

More examples

6. The socioeconomic status of people who reside in Cape Town (upper class, middle class, lower class)
7. The responses by citizens on a 5-point rating scale (where 1=Strongly Disagree, 2=Disagree, 3=Unsure, 4=Agree, 5=Strongly Agree) to the statement: “South Africa should be divided into two time zones”
8. The gender of UCT employees.
9. The maximum temperature recorded in March 2013 (in °C)
0. The rating (excellent, good, fair or poor) given to a particular television program by each of a sample of viewers

Overview of NP tests

Single population tests:

Test	Data Type	Data
Tests for randomness of order	Nominal	Independent observations
Chi-square GoF test	Nominal	Independent observations

Overview of NP tests

Two population tests:

Tests for equality of medians	Data type	Data	Parametric test
Wilcoxon Rank Sum (Mann-Whitney U)	Ordinal or non-normal quant.	Independent samples	t-test
Wilcoxon Signed Rank Sum	Non-normal quant.	Matched/paired samples	matched pairs t-test
Sign Test	Ordinal	Matched/paired samples	matched pairs t-test

Overview of NP tests

Three or more population tests:

Test	Data type	Data	Parametric test
Kruskal-Wallis	Ordinal or non-normal quant.	Independent samples	OW ANOVA
Friedman	Ordinal or non-normal quant.	Matched/blocked samples	TW ANOVA without INT

Overview of NP tests

For relationship between two variables:

Test	Data type	Data	Parametric test
Spearman's rank correlation	Ordinal or non-normal quant.	Paired observations	Pearson's correlation

Ranking Data

Steps:

1. Order data in some sequence (usually ascending)
2. Assign ranks: identify relative position of each value in ordered data
3. Identify ties: when two or more observations have the same value
4. No ties: rank = relative position
5. Ties: assign average rank to tied values

Example 1

- Data: 4, 9, 6, 7, 5, 2, 8

Data	4	9	6	7	5	2	8
Ordered	2	4	5	6	7	8	9
Relative position	1	2	3	4	5	6	7

- No ties, so ranks are as above

Example 2

- Data: 29, 18, 29, 19, 20, 21, 34, 33, 30, 23

Data	29	18	29	19	20	21	34	33	30	23
Ordered	18	19	20	21	23	29	29	30	33	34
Relative position	1	2	3	4	5	6	7	8	9	10

- Ties: 29 and 29 are tied for ranks 6 and 7 → assign **6.5** to both
- Final ranks: 1, 2, 3, 4, 5, 6.5, 6.5, 8, 9, 10

Example 3

- Data: 29, 18, 29, 19, 20, 21, 20, 33, 30, 23, 33, 33, 24

Data	29	18	29	19	20	21	20	33	30	23	33	33	24
Ordered	18	19	20	20	21	23	24	29	29	30	33	33	33
Relative position	1	2	3	4	5	6	7	8	9	10	11	12	13

- Ties:
 - 20 and 20 → ranks 3 & 4 → assign **3.5**
 - 29 and 29 → ranks 8 & 9 → assign **8.5**
 - 33, 33, 33 → ranks 11, 12, 13 → assign **12**
- Final ranks: 1, 2, 3.5, 3.5, 5, 6, 7, 8.5, 8.5, 10, 12, 12, 12

Lecture 2

Recap of lecture 1

1. Parametric techniques are based on the idea that your data comes from a population that has a known or hypothesized distribution.
2. Non-parametric techniques are used for data that does not meet the assumptions of parametric tests, such as non-normality, small sample sizes, or qualitative data.
3. Data types: Qualitative (Nominal, Ordinal) and Quantitative (Interval, Ratio)
4. Ranking data: order data, assign ranks, identify ties, assign average rank to tied.

Learning outcomes

1. Know the conditions (type of data, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Wilcoxon Signed Rank Sum Test.
2. Know the assumptions and objective of the Wilcoxon Signed Rank Sum Test and be able to conduct the test for two matched samples.

Wilcoxon Signed Rank Sum Test

- When is it appropriate?
 - Compare two matched samples of quantitative (interval or ratio) with respect to central locations
 - I.e. determine whether they come from the same population
- Remember: t-tests $\rightarrow$ mean of differences between samples (μ_D)
- Here: we look at the median (central tendency) of differences
- H_0 : median of differences = 0 (i.e. no difference between samples)
- No evidence for a systematic shift in median difference.

Hypotheses

1. H_0 : median of differences = 0 (i.e. no difference between samples)
 2. H_1 : median of differences $\neq 0$ (i.e. there is a difference between samples) OR
 3. H_1 : median of differences > 0 (i.e. sample 1 has higher values than sample 2) OR
 4. H_1 : median of differences < 0 (i.e. sample 1 has lower values than sample 2)
- Note: you are required to state these hypotheses using the information from the question context, i.e. do not simply state these theoretical hypotheses.

Data and assumptions

- Two paired samples
- Quantitative data (interval or ratio)
- Under H_0 , the distribution of the population of differences within pairs is symmetric around the median.
- The n paired differences are independent (and random samples).

Calculating the test statistic

1. Calculate the difference for each pair
2. Exclude pairs with difference = 0 (i.e. no difference)
3. Record n = number of non-zero differences
4. Record the sign of the paired differences
5. Rank the absolute values of the differences (ignoring signs)
6. Reapply the sign to the ranks
7. Calculate the test statistics, W , as the sum of the signed ranks.

The question

We measure the **same people** before and after a treatment.

$$d_i = \text{Before}_i - \text{After}_i$$

Did the treatment do anything?

$$H_0 : \text{median of differences} = 0$$

What does “no effect” look like?

If H_0 is true:

- Some people lose weight, some gain — roughly **equal numbers**
- The **sizes** of gains and losses are similar
- No systematic pattern

We need a test that can detect when this breaks down — **without assuming normality**.

Idea: Work with **ranks** instead — just the ordering, not the exact values.

The procedure

1. Compute $|d_i|$
2. **Rank** from smallest to largest
3. **Reattach** the original sign (+/-)
4. **Sum** the signed ranks $\rightarrow W$

$$W = \sum \text{sign}(d_i) \cdot R_i$$

Why does this work?

Under H_0 , the + and – signs are **random** across the ranks.

Rank 1 might be + or –. Same for rank 6.

The positive and negative ranks roughly **cancel out**.

$$W \approx 0$$

Scenario 1: The placebo

Person	Before	After	d_i
1	80.0	78.6	+1.4
2	73.5	76.0	-2.5
3	85.0	81.2	+3.8
4	69.0	74.1	-5.1
5	77.8	78.5	-0.7
6	90.2	86.0	+4.2

Scenario 1: Signed ranks

$ d_i $	Rank	Sign	Signed Rank
0.7	1	–	–1
1.4	2	+	+2
2.5	3	–	–3
3.8	4	+	+4
4.2	5	+	+5
5.1	6	–	–6

$$W = -1 + 2 - 3 + 4 + 5 - 6 = +1$$

Scenario 1: What do we see?

The signs are **scattered** across small and large ranks.

The **biggest rank (6) is negative** — the largest change was someone who *gained* weight.

$W = +1$, practically **zero**.

No pattern. Exactly what we'd expect if the treatment did nothing.

Scenario 2: The drug works

Person	Before	After	d_i
1	82.5	79.3	+3.2
2	68.0	69.1	-1.1
3	91.2	85.4	+5.8
4	75.0	73.0	+2.0
5	88.3	83.8	+4.5
6	70.1	62.8	+7.3

Scenario 2: Signed ranks

$ d_i $	Rank	Sign	Signed Rank
1.1	1	−	−1
2.0	2	+	+2
3.2	3	+	+3
4.5	4	+	+4
5.8	5	+	+5
7.3	6	+	+6

$$W = -1 + 2 + 3 + 4 + 5 + 6 = +19$$

Scenario 2: What do we see?

The **big ranks** (4, 5, 6) are all positive.

The only negative rank is the **smallest** (rank 1).

$W = +19$ is **far from zero**.

The people with the **largest weight changes** all lost weight. That's not random — that's evidence the drug works.

The key contrast

	Scenario 2 (drug)	Scenario 1 (placebo)
W	+19	+1
Big ranks	Almost all +	Mixed + and –
Pattern?	Systematic	Random
Conclusion	Reject H_0	Do not reject H_0

So what is W really measuring?

W asks: are the signs distributed randomly across the ranks, or is there a pattern?

- If signs are random $\rightarrow W \approx 0 \rightarrow$ no evidence
- If large ranks are mostly $+$ $\rightarrow W \gg 0 \rightarrow$ first sample tends to be larger
- If large ranks are mostly $-$ $\rightarrow W \ll 0 \rightarrow$ second sample tends to be larger

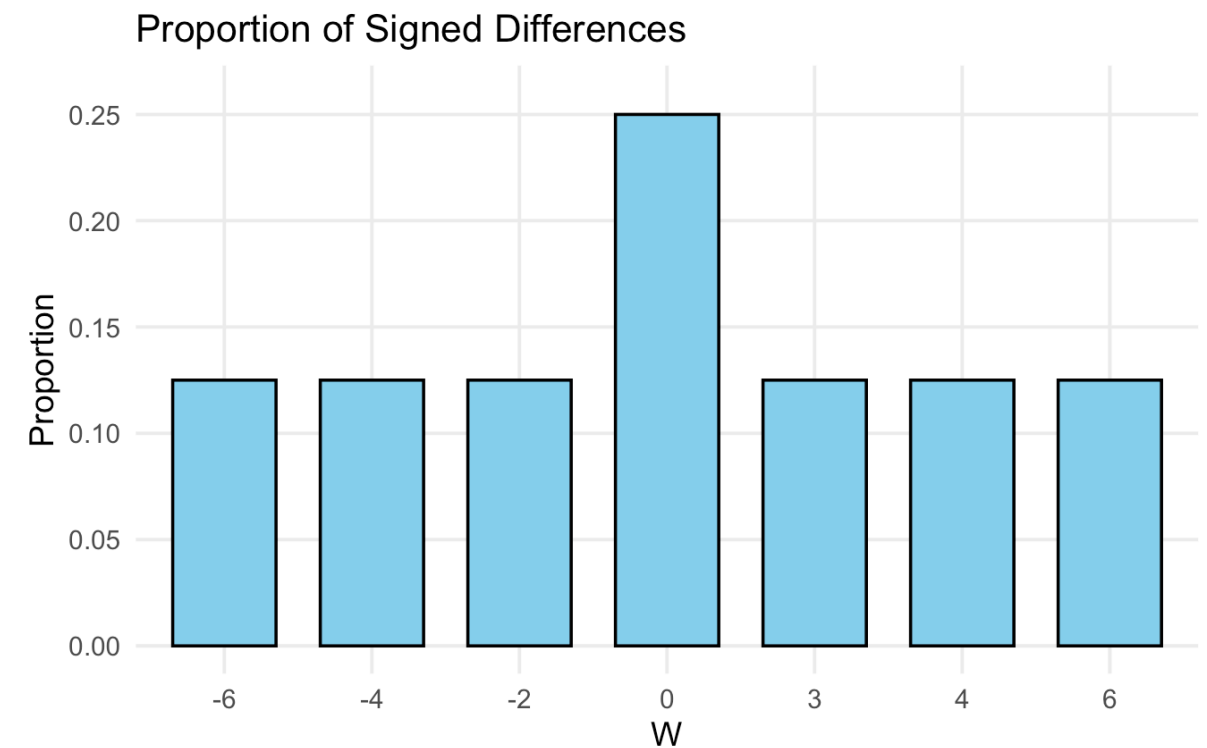
Ask always, we ask, how extreme is the observed W under H_0 ? How different from zero is different enough?

Sampling distribution of W

For small n (with no ties), it is possible to deduce the properties of the sampling distribution of W through simple enumeration of all possibilities

Ranks

1	2	3	W
+	+	+	+6
-	+	+	+4
+	-	+	+3
+	+	-	0
-	-	+	0
-	+	-	-2
+	-	-	-4
-	-	-	-6



Large sample sizes

- For large n ($n > 10$), the sampling distribution of W can be approximated by a normal distribution
 - with mean $\mu_W = 0$ and
 - standard deviation $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}}$
- Critical value approach:
 - Reject H_0 if $|z| > z_{\alpha/2}$ for a two-tailed test.
 - Reject H_0 if $z > z_{\alpha}$ for a right-tailed test.
 - Reject H_0 if $z < -z_{\alpha}$ for a left-tailed test.
- P-value approach:
 - Calculate the p-value corresponding to the observed z -value.
 - Reject H_0 if p-value $< \alpha$.

Example

Does a “flexi-time” work-schedule help reduce the travel time of workers to work?

- A random sample of 32 workers was selected, and workers recorded their travel time in minutes before and after the program was implemented.
- Test at the 5% significance level, using the modified (p-value) approach.

Data

Worker	8:00-Arr	Flextime
1	34	31
2	35	31
3	43	44
4	46	44
5	16	15
6	26	28
7	68	63
8	38	39
9	61	63
10	52	54
11	68	65
12	13	12
13	69	71
14	18	13
15	53	55
16	18	19

Worker	8:00-Arr	Flextime
17	41	38
18	25	23
19	17	14
20	26	21
21	44	40
22	30	33
23	19	18
24	48	51
25	29	33
26	24	21
27	51	50
28	40	38
29	26	22
30	20	19
31	19	21
32	42	38

Hypotheses

- We want to compare two matched/paired samples of quantitative data. The samples do not adhere to a normal distribution,
- and we are interested in whether there is a difference between them, so we use the Wilcoxon Signed Rank Sum test.
- H0: There is no difference in travel time to work between the normal work-hour and flexi-time work programs. The median difference is zero.
- H1: Workers take longer to travel to work in the normal work-hour program i.e. workers take less time to travel to work in the flexi-time program (hence flexitime does reduce workers' travel time). The median difference is greater than zero.

Example 1

Worker	8:00-Arr	Flextime	Diff	Abs Diff
1	34	31	3	3
2	35	31	4	4
3	43	44	-1	1
4	46	44	2	2
5	16	15	1	1
6	26	28	-2	2
7	68	63	5	5
8	38	39	-1	1
9	61	63	-2	2
10	52	54	-2	2
11	68	65	3	3
12	13	12	1	1
13	69	71	-2	2
14	18	13	5	5
15	53	55	-2	2
16	18	19	-1	1

Worker	8:00-Arr	Flextime	Diff	Abs Diff
17	41	38	3	3
18	25	23	2	2
19	17	14	3	3
20	26	21	5	5
21	44	40	4	4
22	30	33	-3	3
23	19	18	1	1
24	48	51	-3	3
25	29	33	-4	4
26	24	21	3	3
27	51	50	1	1
28	40	38	2	2
29	26	22	4	4
30	20	19	1	1
31	19	21	-2	2
32	42	38	4	4

Example 1

Worker	Diff	Abs Diff	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	Abs Diff	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

Example 1

Worker	Diff	Abs Diff	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	Abs Diff	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

8-way tie: assign the average of ranks 1 to 8 to each value:

Average =

$$\frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8}{8}$$

Example 1

Worker	Diff	Abs Diff	Rank
1	3	3	21
2	4	4	27
3	-1	1	4.5
4	2	2	13
5	1	1	4.5
6	-2	2	13
7	5	5	31
8	-1	1	4.5
9	-2	2	13
10	-2	2	13
11	3	3	21
12	1	1	4.5
13	-2	2	13
14	5	5	31
15	-2	2	13
16	-1	1	4.5

Worker	Diff	Abs Diff	Rank
17	3	3	21
18	2	2	13
19	3	3	21
20	5	5	31
21	4	4	27
22	-3	3	21
23	1	1	4.5
24	-3	3	21
25	-4	4	27
26	3	3	21
27	1	1	4.5
28	2	2	13
29	4	4	27
30	1	1	4.5
31	-2	2	13
32	4	4	27

Example 1

Worker	Diff	Abs Diff	Rank	Signed Rank
1	3	3	21	+21
2	4	4	27	+27
3	-1	1	4.5	-4.5
4	2	2	13	+13
5	1	1	4.5	+4.5
6	-2	2	13	-13
7	5	5	31	+31
8	-1	1	4.5	-4.5
9	-2	2	13	-13
10	-2	2	13	-13
11	3	3	21	+21
12	1	1	4.5	+4.5
13	-2	2	13	-13
14	5	5	31	+31
15	-2	2	13	-13
16	-1	1	4.5	-4.5

Worker	Diff	Abs Diff	Rank	Signed Rank
17	3	3	21	+21
18	2	2	13	+13
19	3	3	21	+21
20	5	5	31	+31
21	4	4	27	+27
22	-3	3	21	-21
23	1	1	4.5	+4.5
24	-3	3	21	-21
25	-4	4	27	-27
26	3	3	21	+21
27	1	1	4.5	+4.5
28	2	2	13	+13
29	4	4	27	+27
30	1	1	4.5	+4.5
31	-2	2	13	-13
32	4	4	27	+27

Test statistic

- n = number of non-zero differences = $32 > 10$
- Test Statistic: W = sum of ranked signed paired differences = 207
- $\Rightarrow z = \frac{W - \mu_W}{\sigma_W}$ where $\mu_W = 0$ and $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{32(33)(65)}{6}} = 106.958$
- $\Rightarrow z = \frac{207 - 0}{106.958} = 1.935$
- Critical Region: Reject H_0 if p-value ≤ 0.05
- p-value of test statistic?
- Use R! `pnorm(1.935,lower.tail = F) = [1] 0.02649515`

Conclusion

Since $p\text{-value} = 0.0262 < 0.05$ we Reject H_0 at the 5% significance level and conclude that workers do take longer to travel to work during normal hours than flexitime i.e. there is sufficient evidence that flexi-time does reduce the travel time of workers to work.

Example 2

Example 2

Subject	X_A	X_B	$X_A - X_B$	$ X_A - X_B $	Abs rank	Signed rank
1	78	78	0	0	—	—
2	24	24	0	0	—	—
3	64	62	+2	2	1	+1
4	45	48	-3	3	2	-2
5	64	68	-4	4	3.5	-3.5
6	52	56	-4	4	3.5	-3.5
7	30	25	+5	5	5	+5
8	50	44	+6	6	6	+6
9	64	56	+8	8	7	+7
10	50	40	+10	10	8.5	+8.5
11	78	68	+10	10	8.5	+8.5
12	22	36	-14	14	10	-10
13	84	68	+16	16	11	+11
14	40	20	+20	20	12	+12
15	90	58	+32	32	13	+13
16	72	32	+40	40	14	+14

$W = 67.0, n = 14$

Conclusion

- n = number of non-zero differences = $14 > 10$
- Test Statistic: W = sum of ranked signed paired differences = 67

- $\Rightarrow z = \frac{W - \mu_W}{\sigma_W}$ where $\mu_W = 0$ and $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{14(15)(29)}{6}} = 31.86$

- $\Rightarrow z = \frac{67 - 0}{31.86} = 2.10$

- Critical Region: Reject H_0 if p-value ≤ 0.05
- p-value of test statistic? $\text{pnorm}(2.10, \text{lower.tail} = \text{F}) = 0.0179$

Lecture 3

1. Wilcoxon Signed Rank Sum test, is used for two paired samples of non-normal quantitative data
2. Tests the null hypothesis that the median difference is zero
3. It achieves this by ranking the differences between samples, re-applying sign of the difference to the ranks and calculating the sum of signed ranks as the test statistic.
4. For sample sizes greater than 10, this test statistic is approximately normally distributed so can use a z-score to draw conclusions about null hypothesis.

Learning out comes

1. Know the differences between the Wilcoxon Signed Rank Sum Test and the Mann-Whitney-Wilcoxon Test
2. Know the conditions (data type, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Mann-Whitney-Wilcoxon test
3. Know the data assumptions of the Mann-Whitney-Wilcoxon Test
4. Be able to conduct the Mann-Whitney-Wilcoxon Test

Data and assumptions

Also called the U-test, Wilcoxon Rank Sum Test, or just Rank Sum Test.

- When is it appropriate? The objective of this test is to determine whether two independent samples of ordinal or quantitative data have the same location (median) i.e. are from the same population
- It is the equivalent of the t-test for two samples of normal data.

Data and Assumptions:

- 2 random samples of size n_1 and n_2 .
- The data are either ordinal or quantitative (but not normal)
- Samples and observations within samples are independent.
- The distributions of the two populations differ with respect to location only (if they differ at all).

Hypotheses

- H_0 : The two population locations (medians) are the same / identical distribution
- H_1 : The two population locations (medians) are different / different distribution OR
- H_1 : The location of the first population is to the right of the location of the second population / first population tends to have larger values than the second population OR
- H_1 : The location of the first population is to the left of the location of the second population / first population tends to have smaller values than the second population

Calculating the test statistic

Test stat depends on n_1 and n_2

Steps:

1. Combine the two samples into a single set of values
2. Rank all observations from smallest to largest, that is, from 1 to $n_1 + n_2$
3. Calculate the sum of ranks: T_1 = sum of ranks for sample 1, T_2 = sum of ranks for sample 2

- Note: $T_1 + T_2 = \frac{(n_1+n_2)(n_1+n_2+1)}{2}$

Logic

- If the locations of the two populations are about the same (i.e. the null hypothesis is true)
- we would expect the ranks to be evenly spread between the samples and the sum of the ranks to be similar.
- If T_1 is sufficiently small, then most of the smaller observations are in population 1
 - The location of population 1 is to the left of the location of population 2
 - Reject the null hypothesis
- If T_1 is sufficiently large, then most of the larger observations are in population 1
 - the location of population 1 is to the right of the location of population 2
 - Reject the null hypothesis.

Example

- Sample 1 = 0 1 1 0 1 2 1 2 3
- Sample 2 = 7 9 10 8 10 11 10 11 12

0	1	1	0	1	2	1	2	3	7	9	10	8	10	11	10	11	12
1.5	4.5	4.5	1.5	4.5	7.5	4.5	7.5	9	10	12	14	11	14	16.5	14	16.5	18

- Clearly ranks are not evenly spread
- Sum of ranks for sample 1 = 45
- Sum of ranks for sample 2 = 126

Mann-Whitney-Wilcoxon Test

Small sample Wilcoxon Rank Sum Test (n_1 AND $n_2 < 10$)

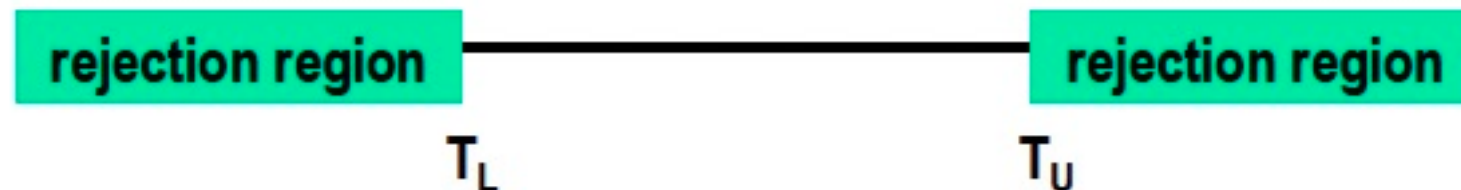
Test statistic: $T = T_1$

Critical Region:

- when n_1 AND $n_2 < 10$, critical values of T_L can be found in the appropriate table. The value of T_U is calculated as:

$$T_U = n_1(n_1 + n_2 + 1) - T_L$$

- Reject H_0 if $T \leq T_L$ or $T \geq T_U$



Mann-Whitney-Wilcoxon Test

Large sample Wilcoxon Rank Sum Test (n_1 AND/OR $n_2 \geq 10$)

Test Statistic: T is approximately normally distributed

$$z = \frac{T - \mu_T}{\sigma_T} \quad \text{where} \quad \mu_T = \frac{n_1(n_1 + n_2 + 1)}{2} \quad \text{and} \quad \sigma_T = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

Critical value approach:

Reject H_0 if $|z| \geq z_{\alpha/2}$ (two-sided test) OR

if $z > z_{\alpha}$ (one-sided > test) OR

if $z < -z_{\alpha}$ (one-sided < test)

where z_{α} is the critical value

p-value value approach:

if the calculated p-value $\leq \alpha$

Example

- The ABC Company has sent 13 of its employees to a privately-run programme providing word-processing skills training. Six of the employees were from the data-processing (DP) department and the others were from the typing (T) pool.
- At the end of the programme, the company received a report indicating the score received by each of its employees out of a total possible score of 100.
- The scores of the 13 employees of ABC are given in the following table (next slide):
- Is there a difference in the performance of the two groups in the word-processing programme? Test at a 5% significance level.

Data

DP	T
70	59
52	70
46	75
65	85
60	50
40	82
	64

Hypothesis and sample sizes

We want to compare two **independent** samples of quantitative data with respect to their locations — use the Mann-Whitney-Wilcoxon test.

- H_0 : There is no difference in performance between the two groups on the word processing program
- H_1 : There is a difference in performance between the two groups on the word processing program
- $n_1 = 6$
- $n_2 = 7$
- $n_1 \text{ AND } n_2 < 10 \implies$ Small Sample Test

Procedure

Calculating the Test Statistic:

DP	T
70	59
52	70
46	75
65	85
60	50
40	82
	64

Combined Sample	Ordered Combined Sample	Rank
70		
52		
46		
65		
60		
40		
59		
70		
75		
85		
50		
82		
64		

Mann-Whitney-Wilcoxon Test — Example 1

Calculating the Test Statistic:

DP	T
70	59
52	70
46	75
65	85
60	50
40	82
	64

Combined Sample	Ordered Combined Sample	Rank
70	40	1
52	46	2
46	50	3
65	52	4
60	59	5
40	60	6
59	64	7
70	65	8
75	70	9.5
85	70	9.5
50	75	11
82	82	12
64	85	13

$$T_1 = 1 + 2 + 4 + 6 + 8 + 9.5 = 30.5$$

$$T_2 = 3 + 5 + 7 + 9.5 + 11 + 12 + 13 = 60.5$$

Example 1: Critical Region

Critical Region: Reject H_0 if $T \leq T_L$ or $T \geq T_U$

TABLE 8 — T_L Values for the Mann–Whitney–Wilcoxon test ($\alpha = 0.05$)

n_1	2	3	4	5	6	7	8	9	10
2	3	3	3	3	3	3	4	4	4
3	6	6	6	7	8	8	9	9	10
4	10	10	11	12	13	14	15	15	16
5	15	16	17	18	19	21	22	23	24
6	21	23	24	25	27	28	30	32	33
7	28	30	32	34	35	37	39	41	43
8	37	39	41	43	45	47	50	52	54
9	46	48	50	53	56	58	61	63	66
10	56	59	61	64	67	70	73	76	79

$n_1 = 6, n_2 = 7: \quad T_L = 28 \qquad T_U = n_1(n_1 + n_2 + 1) - T_L = 6(6 + 7 + 1) - 28 = 56$

Example 1: Conclusion

Critical Region: Reject H_0 if $T \leq 28$ or $T \geq 56$

$$T = T_1 = 30.5$$

Conclusion: Since $28 < T < 56$ we do **NOT Reject** H_0 and we conclude that there is no significant difference in the performance of the two groups in the word-processing programme at the 5% significance level

Example 2: Context

A pharmaceutical company is planning to introduce a new painkiller. To determine the effectiveness of the drug in comparison to aspirin, 30 people were randomly selected.

15 were given the new drug (Sample 1).

15 were given aspirin (Sample 2).

Each participant was asked to indicate which one of five statements best represented the effectiveness of the drug they took.

The drug taken was...

(5) Extremely effective

(4) Quite effective

(3) Somewhat effective

(2) Slightly effective

(1) Not at all effective

Example 2: Data

New	Aspirin
3	4
5	1
4	3
3	2
2	4
5	1
1	3
4	4
5	2
3	2
3	2
5	4
5	3
5	4
4	5

Summary of Results:

The drug's score was...	New	Aspirin
5	6	1
4	3	5
3	4	3
2	1	4
1	1	2

Can we conclude that the new painkiller is perceived to be more effective?

Test at the 5% significance level

Example 2: Hypotheses

We want to compare two independent samples of ordinal data with respect to their locations → use the Mann–Whitney–Wilcoxon test.

H_0 : There is no difference in effectiveness between the two painkillers.

H_1 : The new painkiller is more effective (one-sided $>$ test).

$$n_1 = 15, \quad n_2 = 15$$

n_1 and $n_2 > 10 \Downarrow$ **Large Sample Test**

Example 2: Combined Sample with Ranks

Calculating the Test Statistic:

New	Rank
3	12
5	27
4	19.5
3	12
2	6
5	27
1	2
4	19.5
5	27
3	12
3	12
5	27
5	27
5	27
4	19.5

Aspirin	Rank
4	19.5
1	2
3	12
2	6
4	19.5
1	2
3	12
4	19.5
2	6
2	6
2	6
4	19.5
3	12
4	19.5
5	27

Example 2: Rank Sums

New	Rank
3	12
5	27
4	19.5
3	12
2	6
5	27
1	2
4	19.5
5	27
3	12
3	12
5	27
5	27
5	27
4	19.5
T_1	276.5

Aspirin	Rank
4	19.5
1	2
3	12
2	6
4	19.5
1	2
3	12
4	19.5
2	6
2	6
4	19.5
3	12
4	19.5
5	27
T_2	188.5

Example 2: Test Statistic

Test Statistic: $T = T_1 = 276.5$

$$\implies z = \frac{T - \mu_T}{\sigma_T} \quad \text{where: } \mu_T = \frac{n_1(n_1 + n_2 + 1)}{2} = \frac{15(15 + 15 + 1)}{2} = 232.5$$

$$\sigma_T = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{(15)(15)(15 + 15 + 1)}{12}} = 24.11$$

$$\implies z = \frac{276.5 - 232.5}{24.11} = 1.82$$

Critical Region: Reject H_0 if $z \geq 1.645$ OR approximate p-value ≤ 0.05

Example 2: P-value

Test statistic = 1.82

From the standard normal table:

$$P(Z < 1.82) = 0.9656$$

$$\text{p-value} = 1 - 0.9656 = 0.0344$$

Example 2: Conclusion

p-value ≈ 0.0344

Conclusion (a): Since $1.82 > 1.645$ we **Reject** H_0 and conclude that the new painkiller is more effective at the 5% level of significance

OR

Conclusion (b): Since p-value $\approx 0.0344 < 0.05$ we **Reject** H_0 and conclude that the new painkiller is more effective at the 5% level of significance